Outils Mathématiques – Calcul Différentiel

CD1

1) $\frac{\partial f}{\partial x} = \frac{1}{x-y}\cos(x^2+y^2) + \ln(|x-y|)(-2x\sin(x^2+y^2))$

$\frac{\partial f}{\partial y} = -\frac{1}{x-y}\cos(x^2+y^2) + \ln(|x-y|)(-2y\sin(x^2+y^2))$

Finalement :

$df = [\frac{\cos(x^2+y^2)}{x-y} - \ln(|x-y|)2x\sin(x^2+y^2)]dx + [\frac{-\cos(x^2+y^2)}{x-y} - \ln(|x-y|)2y\sin(x^2+y^2)]dy$

2) Soit $f(x,y) = x\cos y + y\exp(x)$

$\frac{\partial f}{\partial x} = \cos(y) + y\,e^x$ $\qquad$ $\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial f}{\partial y}(\frac{\partial f}{\partial x}) = \frac{\partial f}{\partial y}(\cos(y) + y\,e^x) = -\sin(y) + e^x$

$\frac{\partial f}{\partial y} = -x\sin(y) + e^x$ $\qquad$ $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial f}{\partial x}(-x\sin(y) + e^x) = -\sin(y) + e^x$

$\frac{\partial^2 f}{\partial x^2} = \frac{\partial f}{\partial x}(\cos(y) + y\,e^x) = y\,e^x$ $\qquad$ $\frac{\partial^2 f}{\partial y^2} = \frac{\partial f}{\partial y}(-x\sin(y) + e^x) = -x\cos(y)$

Soit $f(x,y) = x^2\exp(xy)$

$\frac{\partial f}{\partial x} = 2x\,e^{xy} + x^2 y\,e^{xy}$ $\qquad$ $\frac{\partial^2 f}{\partial y \partial x} = 2x^2 e^{xy} + x^2 e^{xy} + x^3 y\,e^{xy} = 3x^2 e^{xy} + x^3 y\,e^{xy}$

$\frac{\partial f}{\partial y} = x^2 x\,e^{xy} = x^3 e^{xy}$ $\qquad$ $\frac{\partial^2 f}{\partial x \partial y} = 3x^2 e^{xy} + x^3 y e^{xy}$

$\frac{\partial^2 f}{\partial y^2} = x^3 x\,e^{xy} = x^4 e^{xy}$ $\qquad$ $\frac{\partial^2 f}{\partial x^2} = 2e^{xy} + 2xy\,e^{xy} + y\,2x\,e^{xy} + yx^2 y\,e^{xy} = 2e^{xy} + 4xy\,e^{xy} + x^2 y^2 e^{xy}$

CD3

1) Soit $f(x,y) = \sin(xy)$

$\frac{\partial f}{\partial x} = y\cos xy$ $\qquad$ $\frac{\partial f}{\partial y} = x\cos xy$

2) $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial x} = (\frac{\partial f}{\partial u})\times 1 + (\frac{\partial f}{\partial v})\times 1 = \frac{\partial f}{\partial u} + \frac{\partial f}{\partial v}$

$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial y} = (\frac{\partial f}{\partial u})\times 1 + (\frac{\partial f}{\partial v})\times(-1) = \frac{\partial f}{\partial u} - \frac{\partial f}{\partial v}$

3) Exprimons y puis y en fonction de u et v :

$$\begin{cases} u = x+y \\ v = x-y \end{cases} \Leftrightarrow \begin{cases} u+v = x+y+x-y \\ u-v = x+y-x+y \end{cases} \Leftrightarrow \begin{cases} 2x = u+v \\ 2y = u-v \end{cases} \Leftrightarrow \begin{cases} x = \frac{u+v}{2} \\ y = \frac{u-v}{2} \end{cases}$$

$f(u,v) = \sin(\frac{u+v}{2}\frac{u-v}{2}) = \sin(\frac{u^2-v^2}{4})$

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$$\frac{\partial f}{\partial x}=\frac{\partial f}{\partial u}+\frac{\partial f}{\partial v}=\frac{2u}{4}\cos\left(\frac{u^2-v^2}{4}\right)+\left(\frac{-2v}{4}\cos\left(\frac{u^2-v^2}{4}\right)\right)=\frac{u-v}{2}\cos\left(\frac{u^2-v^2}{4}\right)=y\cos xy$$
$$\frac{\partial f}{\partial y}=\frac{\partial f}{\partial u}-\frac{\partial f}{\partial v}=\frac{2u}{4}\cos\left(\frac{u^2-v^2}{4}\right)-\left(\frac{-2v}{4}\cos\left(\frac{u^2-v^2}{4}\right)\right)=\frac{u+v}{2}\cos\left(\frac{u^2-v^2}{4}\right)=x\cos xy$$

CD4

Loi de Snell : $n_1\sin i_1=n_2\sin i_2\Leftrightarrow i_2=\arcsin\left(\frac{n_1}{n_2}\sin i_1\right)$

1) Calcul direct : $di_2=\frac{n_1}{n_2}\frac{\cos i_1}{\sqrt{1-\left(\frac{n_1}{n_2}\sin i_1\right)^2}}di_1=\frac{n_1}{n_2}\frac{\cos i_1}{\sqrt{(\cos i_2)^2}}di_1=\frac{n_1\cos i_1}{n_2\cos i_2}di_1$

2) Calcul indirect : $n_1\sin i_1=n_2\sin i_2\Leftrightarrow d(n_1\sin i_1)=d(n_2\sin i_2)$
$\Leftrightarrow n_1 d(\sin i_1)=n_2 d(\sin i_2)$
$\Leftrightarrow n_1\cos(i_1)di_1=n_2\cos(i_2)di_2$
$\Leftrightarrow di_2=\frac{n_1\cos i_1}{n_2\cos i_2}di_1$

donc $di_2\approx 0{,}07712°$
or $\Delta i_1=0{,}1°$ ce qui induit une variat° $\Delta i_2\approx di_2\approx 0{,}07712°$

3) $\sin i_2=\frac{n_1}{n_2}\sin i_1\Rightarrow i_2=37{,}639°$

Valeur exacte de Δi_2

$$n_1\sin(i_1+\Delta i_1)=n_2\sin(i_2+\Delta i_2)\Rightarrow\sin(i_2+\Delta i_2)=\frac{n_1}{n_2}\sin(i_1+\Delta i_1)$$
$$\Rightarrow i_2+\Delta i_2=37{,}716°$$
$$\Rightarrow \Delta i_2=0{,}0770°$$

CD5

1) $\frac{\partial P}{\partial T}=\frac{r}{V-b}e^{-\frac{a}{rTV}}+\frac{rT}{V-b}\left(-\frac{a}{rV}\right)\left(-\frac{1}{T^2}\right)e^{-\frac{a}{rTV}}=\frac{r}{V-b}e^{-\frac{a}{rTV}}\left(1+\frac{a}{rVT}\right)$

$\frac{\partial P}{\partial V}=\frac{-rT}{(V-b)^2}e^{-\frac{a}{rTV}}+\frac{rT}{V-b}\left(-\frac{a}{rT}\right)\left(-\frac{1}{V^2}\right)e^{-\frac{a}{rTV}}=\frac{rT}{V-b}e^{-\frac{a}{rTV}}\left(\frac{a}{rTV^2}-\frac{1}{V-b}\right)$

$dP=\left[\frac{r}{V-b}e^{-\frac{a}{rTV}}\left(1+\frac{a}{rVT}\right)\right]dT+\left[\frac{rT}{V-b}e^{-\frac{a}{rTV}}\left(\frac{a}{rTV^2}-\frac{1}{V-b}\right)\right]dV$

2a) $\beta=\frac{1}{P}\frac{\partial P}{\partial T}=\frac{V-b}{rT}\frac{1}{e^{-\frac{a}{rTV}}}\frac{r}{V-b}e^{-\frac{a}{rTV}}\left(1+\frac{a}{rVT}\right)=\frac{1}{T}\left(1+\frac{a}{rVT}\right)$

b) On ne connaît pas $\frac{\partial V}{\partial T}$ (ni $\frac{\partial V}{\partial P}$ pour la qs 2c))

$dV=\frac{\partial V}{\partial P}dP+\frac{\partial V}{\partial T}dT$ (1) et $dP=\frac{\partial P}{\partial T}dT+\frac{\partial P}{\partial V}dV\Leftrightarrow\frac{\partial P}{\partial V}dV=dP-\frac{\partial P}{\partial T}dT$

$$\Leftrightarrow dV=\frac{1}{(\partial P/\partial V)}dP-\frac{(\partial P/\partial T)}{(\partial P/\partial V)}dT \quad (2)$$

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Donc de (1) et (2), il résulte que : $\left\{\begin{array}{l}\dfrac{\partial V}{\partial P}=\dfrac{1}{\partial P/\partial V}\\ \dfrac{\partial V}{\partial T}=\dfrac{-\partial P/\partial T}{\partial P/\partial V}\end{array}\right\}$

Donc $\alpha=\dfrac{1}{V}\dfrac{\partial V}{\partial T}=-\dfrac{1}{V}\dfrac{\dfrac{r}{V-b}e^{-\frac{a}{rTV}}\left(1+\dfrac{a}{rTV}\right)}{\dfrac{rT}{V-b}e^{-\frac{a}{rTV}}\left(\dfrac{a}{rTV^2}-\dfrac{1}{V-b}\right)}=-\dfrac{1}{V}\dfrac{\dfrac{rTV+a}{rTV}}{T\dfrac{a(V-b)-rTV^2}{rTV^2(V-b)}}=-\dfrac{1}{V}\dfrac{1}{T}\dfrac{rTV+a}{rTV}\dfrac{rTV^2(V-b)}{a(V-b)-rTV^2}$

$\alpha=-\dfrac{(rTV+a)(V-b)}{aT(V-b)-rT^2V^2}$

c) $\chi_T=-\dfrac{1}{V}\dfrac{\partial V}{\partial P}=-\dfrac{1}{V}\dfrac{1}{\dfrac{rT}{V-b}e^{-\frac{a}{rTV}}\left(\dfrac{a}{rTV^2}-\dfrac{1}{V-b}\right)}=-\dfrac{1}{V}\dfrac{V-b}{rT}e^{\frac{a}{rTV}}\dfrac{rTV^2(V-b)}{a(V-b)-rTV^2}$

$\chi_T=-\dfrac{V(V-b)^2}{a(V-b)-rTV^2}e^{\frac{a}{rTV}}$

CD8

$df=(3y^2-y^3-x^2)dx+3xy(2-y)dy \quad \Leftrightarrow \left\{\begin{array}{ll}\dfrac{\partial f}{\partial x}=3y^2-y^3-x^2 & (1)\\ \dfrac{\partial f}{\partial y}=3xy(2-y) & (2)\end{array}\right\}$

(1) : $f(x,y)=3xy^2-xy^3-\dfrac{x^3}{3}+G(y)$

$\dfrac{\partial f}{\partial y}=6xy-3xy^2+\dfrac{dG}{dy}=3xy(2-y)+\dfrac{dG}{dy}$, donc d'après (2), $\dfrac{dG}{dy}=0\Leftrightarrow G(y)=C$

or f(1,0) = 1

donc $3\times1\times0^2-1\times0^3-\dfrac{1^3}{3}+C=1 \quad \Leftrightarrow -\dfrac{1}{3}+C=1\Leftrightarrow C=\dfrac{4}{3}$

Finalement : $f(x,y)=3xy^2-xy^3-\dfrac{x^3}{3}+\dfrac{4}{3}$

CD9

> $\delta Q=C_V dT+\dfrac{RT}{V}dV=A\,dT+B\,dV$

$\dfrac{\partial A}{\partial V}=0$ et $\dfrac{\partial B}{\partial T}=\dfrac{R}{V}\neq\dfrac{\partial A}{\partial V}$ donc NON

> $\dfrac{\delta Q}{T}=\dfrac{C_V}{T}dT+\dfrac{R}{V}dV=A\,dT+B\,dV$

$\dfrac{\partial A}{\partial V}=0$ et $\dfrac{\partial B}{\partial T}=0=\dfrac{\partial A}{\partial V}$ donc OUI $\quad \exists f/\dfrac{\delta Q}{T}=df\Rightarrow\left\{\begin{array}{ll}A(T,V)=\dfrac{\partial f}{\partial T}=\dfrac{C_V}{T} & (1)\\ B(T,V)=\dfrac{\partial f}{\partial V}=\dfrac{R}{V} & (2)\end{array}\right\}$

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(1) : $f(T,V)=C_V\ln(T)+G(V)$

$\frac{\partial f}{\partial V}=\frac{dG}{dV}$, donc d'après (2), $\frac{dG}{dV}=\frac{R}{V}\Leftrightarrow G(V)=R\ln(V)+C$

Finalement : $f(T,V)=C_V\ln(T)+R\ln(T)+C$

CD11

$\delta W=F(\sin(\varphi)d\rho+\rho\cos(\varphi)d\varphi)=Fsin(\varphi)d\rho+F\rho\cos(\varphi)d\varphi=A\,d\rho+B\,d\varphi$

$\frac{\partial A}{\partial\varphi}=Fcos\varphi$ et $\frac{\partial B}{\partial\rho}=Fcos\varphi=\frac{\partial A}{\partial\varphi}$ donc OUI, $\exists f/\delta W=df\Rightarrow\left\{\begin{array}{ll}A(\rho,\varphi)=\frac{\partial f}{\partial\rho}=Fsin(\varphi) & (1)\\ B(\rho,\varphi)=\frac{\partial f}{\partial\varphi}=F\rho\cos(\varphi) & (2)\end{array}\right\}$

(1) : $f(\rho,\varphi)=F\rho\sin(\varphi)+G(\varphi)$

$\frac{\partial f}{\partial\varphi}=F\rho\cos(\varphi)+\frac{dG}{d\varphi}$, donc d'après (2), $\frac{dG}{d\varphi}=0\Leftrightarrow G(\varphi)=C$

Finalement : $f(\rho,\varphi)=F\rho\sin(\varphi)+C$

Or f = - E_p , donc $E_p(\rho,\varphi)=-F\rho\sin(\varphi)+C'$

CD13

1) $\frac{1}{R}=\frac{1}{2200}+\frac{1}{120}=\frac{29}{3300}\Rightarrow R=\frac{3300}{29}\,\Omega\approx 113{,}7931034\,\Omega$

2) Calcul direct

$\frac{1}{R}=\frac{1}{R_1}+\frac{1}{R_2}\Leftrightarrow R=\frac{R_1R_2}{R_1+R_2}$; $\Delta R_1=2200\times\frac{15}{100}=330\,\Omega$; $\Delta R_1=120\times\frac{15}{100}=18\,\Omega$

$\frac{\partial R}{\partial R_1}=\frac{R_2(R_1+R_2)+R_1R_2}{(R_1+R_2)^2}=\frac{R_2^2}{(R_1+R_2)^2}$ et $\frac{\partial R}{\partial R_2}=\frac{R_1^2}{(R_1+R_2)^2}$

$\Delta R=\frac{R_2^2}{(R_1+R_2)^2}\Delta R_1+\frac{R_1^2}{(R_1+R_2)^2}\Delta R_2\approx 17{,}06896\,\Omega$

Finalement : $\Delta R\leqslant 20\,\Omega$

donc $R=(110\pm 20)\Omega$

Rmq : il est acceptable d'écrire $\Delta R=17\,\Omega$

$R=(114\pm 17)\Omega$

Prochain TD :

- CD13, 2), avec les différentielles logarithmiques
- CD14
- CD15
- CD16

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